

Notes Spring 2025
TTK 4135 Optimization and Control

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1 Week 3

1.1 Assignments

- Exercises, 7/10 approval required
- Helicopter lab
- Matlab assessments

1.2 Grading

- Final exam: 80%
- Matlab assessments: 20%

1.3 Course material

- Numerical Optimization, J.Nocedal and S. J Wright, 2nd ed.
- Errata on BB.
- Note on Merging Optimization and Control, B. Foss and T.A. Heirung.
- Note on Matrix Calculus, T.A. Heirung.

1.4 Motivation

Optimization: Find best solution to a given task (mathematical problem).
For the non general problems numerical techniques must be used. Applied
mathematics built on:

- Convexity and non-convexity
- Global vs. local solution
- Constrained vs. unconstrained problems
- Feasible set

Typical application for Optimization:

- Autonomous vehicles
- Power plants
- Oil rigs
- Machine learning
- Finance

Minimization and Maximization, convention this course is for minimization
 Maximizing $f(x)$ will be seen as minimizing $-f(x)$.

$$\min_{x \in \mathbb{R}} f(x) \quad (1)$$

What characterizes an optimum?

- $\min_{x \in \mathbb{R}^n} f(x)$
- $f \in \mathbb{C}^k$
- $\frac{d}{dx} f(x^*) = 0$
- $\frac{d^2}{dx^2} f(x^*) > 0$

To make use of optimization, must define an objective which is a quantitative measure of the performance of a system:

- profit
- time
- energy
- or any specific quantity

Objectives depends on variables, and the goal is to find values of the variables that optimize the objective. The variables are often constrained or bounded by some way.

Modeling defines formulating an objective, variables and constraints for a given task. There after an optimization algorithm can be defined, for example in machine learning minimizing MSE.

- x is the vector of variables
- f is the objective function of x returning a scalar which we want to minimize
- c_i are constraint functions of x returning a scalar and defines equations or inequalities that the unknown vector x must satisfy.

1.5 (Un)Constrained Optimization

Constraints are mathematical inequalities for the variables x_n for the desired task.

Unconstrained:

•

$$\min_{x \in \mathbb{R}^n} f(x)$$

Constrained:

•

$$\min_{x \in \mathbb{R}^n} f(x) \text{ s.t. } C_i(x) = 0, i \in \epsilon, C_i(x) \geq 0, i \in \mathbb{I}$$

.

•

$$C_i(x)$$

constrained function

- ϵ : Equality constraint
- I : Inequality constraint

Feasible set:

•

$$\Omega = \{x \in \mathbb{R}^n \text{ s.t. } C_i(x) = 0, i \in \epsilon, C_i(x) \geq 0, i \in \mathbb{I}\}$$

1.6 Linear programming

$$\min_{x \in \mathbb{R}^n} c^T x \text{ s.t. } \begin{cases} a_i^T x = b_i, i \in \epsilon \\ a_i^T x \geq b_i, i \in I \end{cases} \quad (2)$$

1.6.1 Farmer example, find max profit or min neg profit

1.7 Quadratic programming

$$\min_x \frac{1}{2} x^T G x + c^T x \text{ s.t. } \begin{cases} a_i^T x = b_i, i \in \epsilon \\ a_i^T x \geq b_i, i \in I \end{cases} \quad (3)$$

1.8 Convexity

Convex set: A set is convex if

$$x, y \in \mathbb{S} \text{ if } \alpha x + (1 - \alpha)y \in \mathbb{S}, \forall \alpha \in [0, 1] \quad (4)$$

Convex function: $f: \mathbb{S} \rightarrow \mathbb{R}$ is a convex function if $\mathbb{S}$ is convex and *every* $x, y \in \mathbb{S} : f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y), \forall \alpha \in [0, 1]$ "A line from point x to y should be below the function from $f(x)$ to $f(y)$ "

An optimization problem is convex if

- $f(x)$ is a convex function
- The feasible set is a convex set

1.8.1 Local and global solutions

x^* is a global solution if $f(x^*) \leq f(x) \forall x \in \Omega$. Let N be a neighborhood of x : x^* is a local solution if $f(x^*) \leq f(x) \forall x \in N \subseteq \Omega$

Ω is the set of all x values which fulfill the constraints.

2 Week 3

2.1 Gradient

$$\nabla f = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2} \right]$$

2.2 Hessian

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & \frac{\partial^2 f}{\partial x_2 \partial x_n} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} \quad (5)$$

2.3 Lagrangian

For constrained optimization problems: $\mathcal{L}(x, \lambda) = f(x) - \sum_{i \in \mathcal{E} \cup \mathcal{I}} \lambda_i c_i(x)$

2.4 KKT Conditions

If x^* is a local solution and LICQ holds, then there exist λ^* such that:

- $\nabla_x \mathcal{L}(x^*, \lambda^*) = 0$
- $c_i(x^*) = 0, \forall i \in \mathcal{E}$
- $c_i(x^*) \geq 0, \forall i \in \mathcal{I}$

2.5 Proof multipliers are shadow prices

Consider constrained optimization problem: $\min_x f(x) \text{ s.t. } c(x) \geq 0$ KKT stationarity: $\nabla f(x^*) = \lambda^* \nabla c(x^*)$ assume constraint active, $c(x^*) = 0$. Relax constraint: $\min_x f(x) \text{ s.t. } c(x) \geq -\epsilon, \epsilon > 0$ New solution: $x^*(\epsilon)$ with $c(x^*(\epsilon)) = -\epsilon$

$$f(x^*(\epsilon)) - f(x^*) = \nabla f(x^*)^T (x^*(\epsilon) - x^*) = \lambda^* \nabla c(x^*)^T (x^*(\epsilon) - x^*) = \lambda^* (c(x^*(\epsilon)) - c(x^*)) = -\lambda^* \epsilon$$

2.6 Tangent cone

The tangent cone to a set Ω at a point $x \in \Omega$, denoted by $T_\Omega(x)$, consist of the limits of all secant rays which originate at x and pass through a range of points $p_i \in \Omega - x$ which converges to x . A Tangent Cone is a geometric description of the feasible directions

2.7 Active set

$\mathcal{A}(x) = \mathcal{E} \cup \{i \in \mathcal{I} \mid c_i(x) = 0\}$ The active set contains indices of the constraints which are equal 0 for a given point x .

2.8 Linearized feasible directions

Given a feasible point x and the active constraint set $\mathcal{A}(x)$, the set of linearized feasible directions $\mathcal{F}(x)$ is

$$\mathcal{F}(x) = \begin{cases} d^T \nabla c_i(x) = 0, \forall i \in \mathcal{E} \\ d^T \nabla c_i(x) = 0, \forall i \in \mathcal{A}(x) \cap \mathcal{I} \end{cases} \quad (6)$$